Reproduction of Table I of arXiv:1708.02243

S. Knapen, H. K. Lou, M. Papucci, J. Setford, “Tracking down quirks at the Large Hadron Collider”, Phys. Rev. D 96, 115015 (2017) · signal efficiencies, backgrounds and Figures 2–5 · 2000-event samples · 8 October 2026

Summary

- **Trigger and fiducial:** colored trigger 207/873 = 0.24 and fiducial 22/68 = 0.32 agree with the paper (0.24, 0.28). The EW trigger 186/1045 = 0.18 is far above the paper's 0.10; the EW fiducial 62/186 = 0.33 agrees with 0.28.
- **Reconstruction with the paper's SCT z resolution (1.16 mm):** colored 1/22, EW 10/62, far below the paper (0.35, 0.72).
- **Reconstruction without SCT z smearing:** colored 8/22 = 0.36 and EW 50/62 = 0.81, consistent with the paper. The paper's own Fig. 5 and its Λ column also point to this reading.
- **Backgrounds** (cited detector model [34]): 0 of 9139 converted photons + pile-up and 0 of 5000 pile-up events in the signal region, as in the paper.
- **Figures 2–5** have the paper's shapes; the signal strip width Δw is about 2× (EW) and 5× (colored) wider than in the paper's Fig. 5, which stays unexplained.

Verdicts use the exact binomial p-value against the paper's number, taken as exact: $p < 0.01$ strongly inconsistent; 0.01–0.05 statistically inconsistent; 0.05–0.10 borderline / mild tension; $p \geq 0.10$ no statistically significant inconsistency. Intervals are Wilson 95%.

1. Simulation setup

1.1 Event generation (both samples generated with the same setup)

Setting	This work	Paper
Generator	MadGraph5_aMC@NLO 3.7.3 + Pythia 8.317	MadGraph5_aMC@NLO + Pythia8 (source comments: v2.3.3, 8.212)
Processes	colored: p p → t1 t1~ (+ j); EW: p p → qe+ qe− (+ j)	vector-like quirks, up to 1 jet; colored QCD only; EW Drell–Yan, unit charge
Masses, Λ	1800 GeV / 2 keV; 800 GeV / 4 keV	same benchmarks
Beams, PDF, scale	pp 13 TeV; NNPDF2.3 LO (230000); MG5 default dynamical scale	not stated
Matching	MLM, ickkw = 1, xqcut = 30 GeV, qCut = 45 GeV, nJetMax = 1	“matched using Pythia8”, settings not stated
Shower, hadronisation	Pythia 8 defaults (MPI on); colored: R-hadrons on, stable	colored quirks hadronise into quirk-hadrons
Events, seeds	2000 each; MG5 iseed 8001 (colored), 8002 (EW)	not stated

1.2 Cross sections and event statistics

	Colored 1.8 TeV	EW 800 GeV
σ, MadGraph (0 + 1 jet, before matching)	0.753 ± 0.004 fb	0.454 ± 0.002 fb
σ after MLM matching	0.329 ± 0.009 fb	0.237 ± 0.005 fb
Events generated	2000	2000
Events after matching	873	1045
pT(QQ) > 200 GeV (trigger)	207	186
both quirk-hadrons charged (colored)	68	186 (unit charge)
pT(QQ), all matched events: median [16–84%] (GeV)	77 [19–272]	60 [15–223]
pT(QQ), triggered events (GeV)	342 [232–648]	328 [234–575]
M(QQ), all matched events (GeV)	4085 [3775–4660]	1934 [1717–2409]

The trigger is the paper's pT(QQ) > 200 GeV on the final-state quirk pair; for the colored sample, the final-state R-hadrons.

1.3 Detector model and efficiency definitions

Item	This work	Source
Layers	IBL + 3 pixel + 4 SCT barrel layers (8), B = 2 T	paper Sec. III C (8 layers); IBL radius and length inferred (arXiv:1803.00844)
Barrel lengths	IBL $\|z\| < 33.0$ cm, pixel $\|z\| < 40.05$ cm, SCT $\|z\| < 74.9$ cm	Ref. [27]
Pixel hits	pixel centre, 50 × 400 μm (IBL 50 × 250 μm); neighbouring pixels merged	paper Sec. III A
SCT R-φ	34 μm (2 × 17 μm)	paper Sec. III A + Ref. [27]
SCT z (signal)	**1.16 mm (2 × 580 μm)** and **no smearing**, both reported	see Sec. 1.4

Item	This work	Source
Beam spot	45 mm Gaussian, one common shift per event; plane fit from the true vertex	paper Sec. III A, III C
Hits	one hit per quirk per layer (first crossing), t < 25 ns	paper Sec. III B, IV
Plane finding	seeds, windows, factor-3 rule, Δw < 1 cm, smallest Δs; Δs = $\sqrt{\lambda_1}$, Δw = $\sqrt{\lambda_2}$ of Eq. 11	paper Sec. III C
Signal region	plane; 2 hits in 7 layers and ≥ 1 in the 8th; Δw < 1 cm; Δs < 0.01 cm	paper Sec. IV

ε_{trig} = N(pT(QQ̄) > 200 GeV) / N(after matching) ε_{fid} = N(both quirks cross all 8 layers inside the lengths, t < 25 ns) / N(triggered; colored: both charged) ε_{reco} = N(signal region) / N(fiducial) $\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$ (paper Eq. 12)

1.4 Why these SCT z values

Paper, Sec. III A

"for the silicon microstrip tracker (SCT) we conservatively assume a resolution of twice the intrinsic accuracy quoted in [27]."

Ref. [27], ATL-PHYS-PUB-2009-002 (in arXiv:0901.0512, p. 18)

"The intrinsic accuracies per module in the barrel are 17 μm (R−φ) and 580 μm (z)"

- **1.16 mm** is the paper's sentence applied to z: 2 × 580 μm. This is the main setting of this report.
- **No smearing** is what the paper's results need: its Fig. 5 signal Δs medians are 32 μm (colored) and 28 μm (EW); with 1.16 mm we get 160 and 110 μm, without smearing 38 and 44 μm.
- **Backgrounds** use the detector the paper cites for them: pile-up is "processed by the simplified detector described in [34]", and [34] takes its resolutions "from the ATLAS technical design report" (SCT z 580 μm) with energy loss in the material. A comment in the paper's source says this simulation is "somewhat more sophisticated than that for the signal". The backgrounds are therefore the same for both signal settings.

2. Table I: signal efficiencies

2.1 Event counts (2000-event samples, one hit per quirk per layer)

Step	Colored, 1.16 mm	Colored, no smear	EW, 1.16 mm	EW, no smear
after matching	873	873	1045	1045
triggered	207	207	186	186
both charged	68	68	—	—
fiducial	22	22	62	62
plane found (Δw < 1 cm)	9	18	45	62
+ hit multiplicity	4	8	24	50
+ Δs < 0.01 cm (signal region)	1	8	10	50
signal region, every crossing as a hit (≥ 2 hits)	2	11	12	49

Quirks often cross a layer more than once within 25 ns; the paper expects one crossing. The italic row counts every crossing as a hit (sensitivity). 2 colored and 1 EW triggered events stopped with a numerical integration error; they are kept in the ε_{fid} denominator as not fiducial.

2.2 Efficiencies against the paper

Benchmark	Efficiency	k / n	ε [Wilson 95%]	Paper	p	Verdict
Colored 1.8 TeV, 2 keV	ε_{trig}	207 / 873	0.237 [0.210, 0.266]	0.24	0.87	No significant inconsistency
	ε_{fid}	22 / 68	0.324 [0.224, 0.442]	0.28	0.42	No significant inconsistency
	ε_{reco}, SCT z 1.16 mm	1 / 22	0.045 [0.008, 0.218]	0.35	0.001	Strongly inconsistent
	ε_{reco}, no SCT z smearing	8 / 22	0.364 [0.197, 0.570]	0.35	1.00	No significant inconsistency
	ε (Eq. 12)		1.16 mm: 0.0035 no smear: 0.0279	0.024		
EW 800 GeV, 4 keV	ε_{trig}	186 / 1045	0.178 [0.156, 0.202]	0.10	1.8×10^{-14}	Strongly inconsistent
	ε_{fid}	62 / 186	0.333 [0.270, 0.404]	0.28	0.12	No significant inconsistency
	ε_{reco}, SCT z 1.16 mm	10 / 62	0.161 [0.090, 0.272]	0.72	7.8×10^{-20}	Strongly inconsistent
	ε_{reco}, no SCT z smearing	50 / 62	0.806 [0.691, 0.886]	0.72	0.16	No significant inconsistency
	ε (Eq. 12)		1.16 mm: 0.0096 no smear: 0.0478	0.020		

Colored: the both-charged requirement (68 of 207 triggered events) is applied before ε_{fid}, so it is not part of Eq. 12, as in the paper's table.

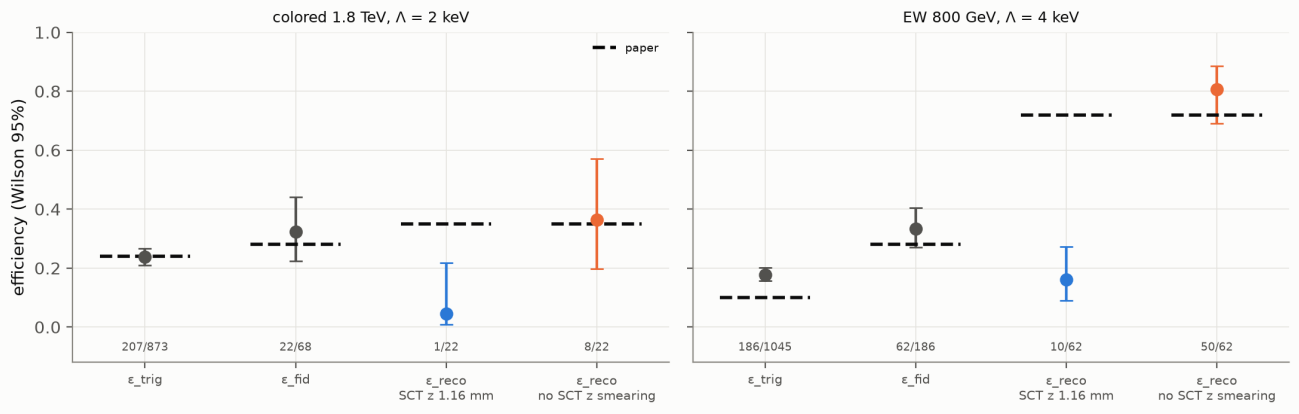
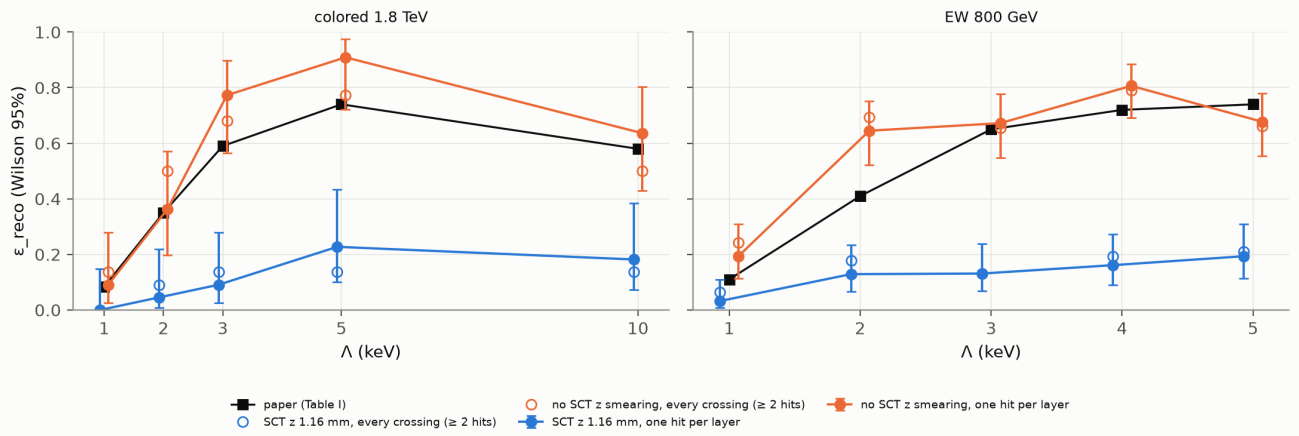


Table I efficiencies with Wilson 95% intervals (points) and the paper's values (dashes). Numbers: k/n.

2.3 The Λ column of Table I

The paper also gives ε_{reco} at other string tensions. The oscillation size scales as $1/\Lambda^2$, so this column tests the reconstruction over a wide range of hit patterns. Trigger and fiducial selections do not depend on Λ in the paper; here the fiducial count is recomputed at every Λ.



ε_{reco} vs Λ. Filled points: one hit per quirk per layer, Wilson 95%. Open points: every crossing as a hit (≥ 2 hits per layer). Blue: SCT z 1.16 mm; orange: no SCT z smearing; black: paper.

Benchmark	Λ (keV)	Paper	1.16 mm: k / n	ε [Wilson 95%]	p	no smear: k / n	ε [Wilson 95%]	p
Colored	1	0.083	0 / 22	0.00 [0.00, 0.15]	0.25	2 / 22	0.09 [0.03, 0.28]	0.70
	2	0.35	1 / 22	0.05 [0.01, 0.22]	0.001	8 / 22	0.36 [0.20, 0.57]	1.00
	3	0.59	2 / 22	0.09 [0.03, 0.28]	1.5×10^{-6}	17 / 22	0.77 [0.57, 0.90]	0.087
	5	0.74	5 / 22	0.23 [0.10, 0.43]	7.3×10^{-7}	20 / 22	0.91 [0.72, 0.97]	0.088
	10	0.58	4 / 22	0.18 [0.07, 0.39]	2.7×10^{-4}	14 / 22	0.64 [0.43, 0.80]	0.67
EW	1	0.11	2 / 62	0.03 [0.01, 0.11]	0.063	12 / 62	0.19 [0.11, 0.31]	0.042
	2	0.41	8 / 62	0.13 [0.07, 0.23]	1.9×10^{-6}	40 / 62	0.65 [0.52, 0.75]	2.4×10^{-4}
	3	0.65	8 / 61	0.13 [0.07, 0.24]	7.0×10^{-17}	41 / 61	0.67 [0.55, 0.78]	0.79
	4	0.72	10 / 62	0.16 [0.09, 0.27]	7.8×10^{-20}	50 / 62	0.81 [0.69, 0.89]	0.16
	5	0.74	12 / 62	0.19 [0.11, 0.31]	3.6×10^{-19}	42 / 62	0.68 [0.55, 0.78]	0.25

Shading of the p column follows the verdict scale (green: no significant inconsistency).

3. Backgrounds

Sample	Detector	Events	Plane	+ hit mult.	Signal region	Paper
converted γ + μ = 50 pile-up	[34] model, SCT z 580 μm	9139	3399	419	0 (95% UL 3.3×10^{-4})	“a few” after the cut
pile-up only, μ = 50	[34] model, SCT z 580 μm	5000	—	—	0 (95% UL 6.0×10^{-4})	none

Pile-up plane rates	This work [Wilson 95%]	Paper	Comparison
selected plane with ≥ 1 hit in ≥ 4 of 8 layers	33/5000 = 6.6×10^{-3} [4.7×10^{-3}, 9.3×10^{-3}]	$\sim10^{-3}$	higher than the paper's approximate rate
selected plane with ≥ 1 hit in all 8 layers	1/5000 = 2.0×10^{-4} [3.5×10^{-5}, 1.1×10^{-3}]	6×10^{-5}	compatible with the paper's approximate rate

Converted photons: Z(→νν)γ + jet, MET > 200 GeV, pT(γ) > 0.5 GeV, σ = 0.211 pb in our sample (paper “~1 pb”), conversion probability 1% taken from the paper's 1 pb → 10 fb. Sensitivity, same events: with SCT z 1.16 mm and no material, 0/9139 conversions and 31/5000 pile-up 4-of-8 planes; with the background also unsmeared, 146/9139 conversions enter the signal region. With an inelastic (instead of non-diffractive) minimum-bias sample the pile-up 4-of-8 rate is 12/5000 = 2.4×10^{-3}.

4. Figures 2–5 of the paper

Paper curves and histograms are read from the vector figures in the paper's arXiv source, so the comparison uses the paper's exact values.

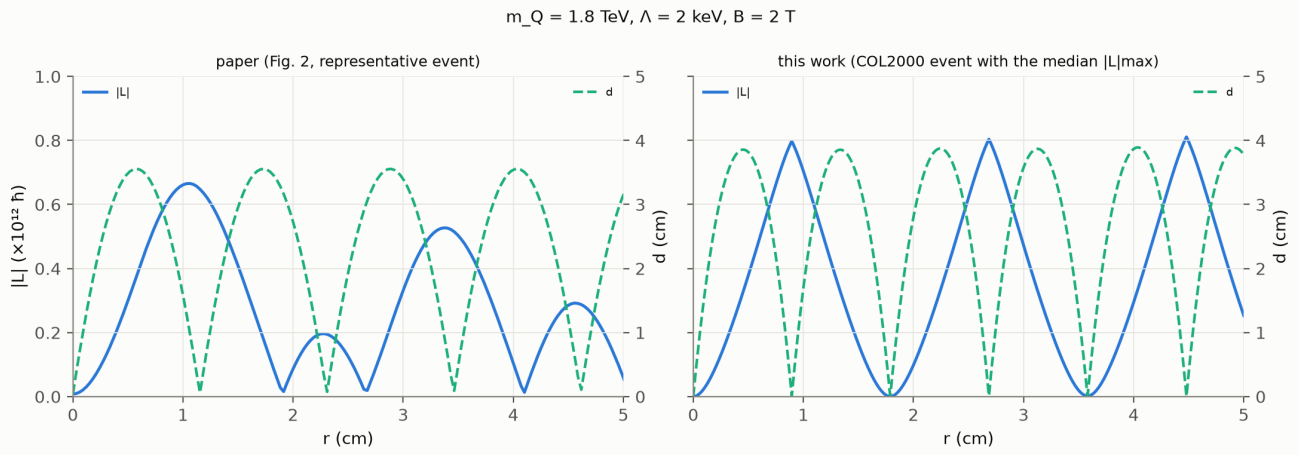


Fig. 2: |L| and quirk separation d vs r, colored 1.8 TeV at 2 keV, exact centre-of-mass frame. Paper event: |L|max = 0.67×10^{12} ħ, d max = 3.56 cm. Our median event: |L|max = 0.81×10^{12} ħ, d max = 3.89 cm. Where |L| ≈ 0, d stays ≥ 0.017 cm ≫ $1/m_Q$, so the pair does not annihilate.

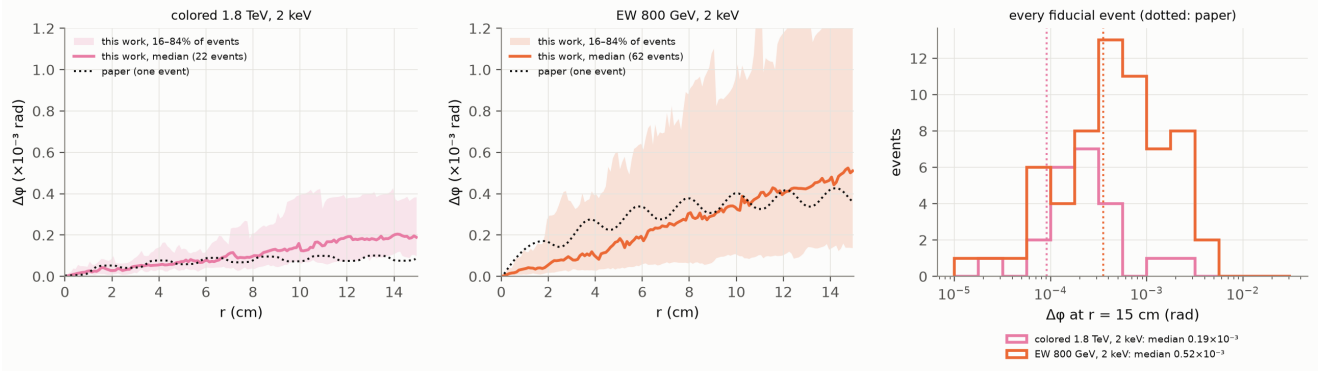


Fig. 3: rotation of the plane spanned by the two velocity vectors (the paper's definition) vs r, both benchmarks at 2 keV. Δφ at 15 cm: colored median 0.19×10^{-3} (paper 0.09×10^{-3}, at our 9th percentile); EW median 0.52×10^{-3} (paper 0.36×10^{-3}, 39th percentile). Both stay far below the tracker resolution.

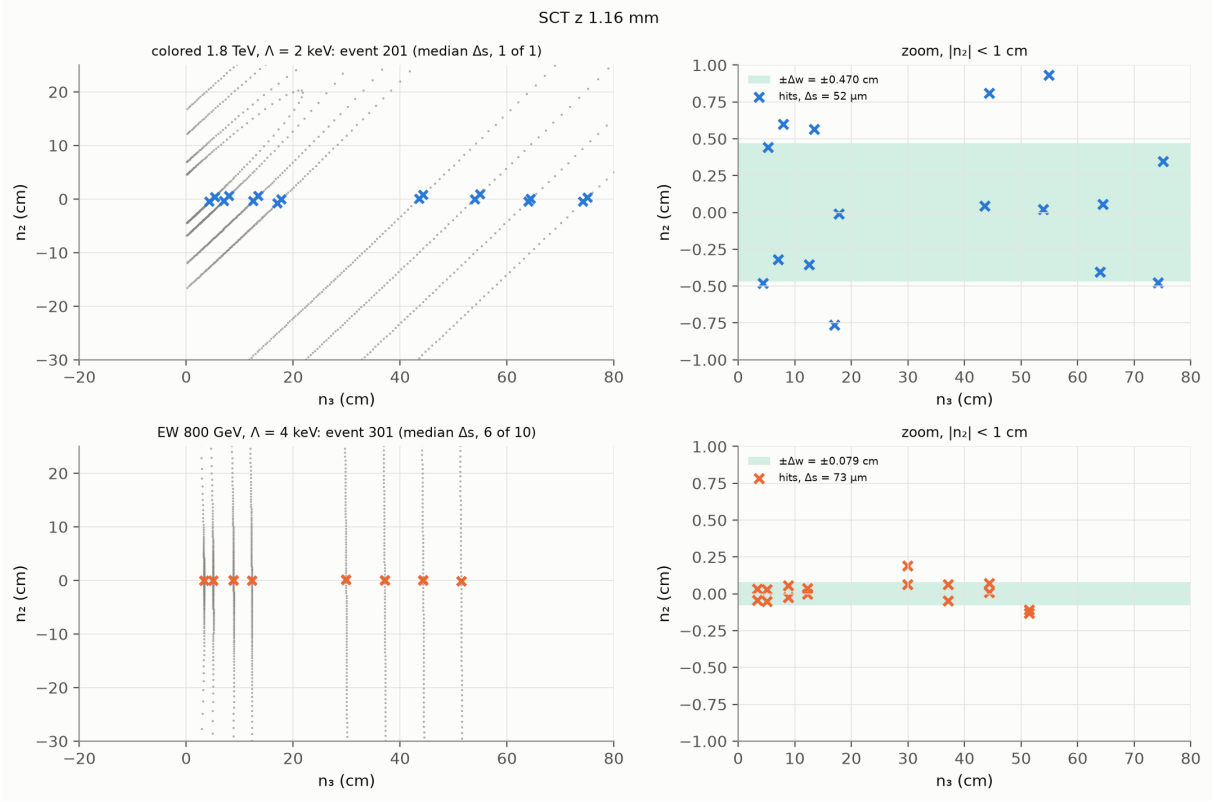


Fig. 4, SCT z 1.16 mm. Fixed selection: the signal-region event with the median Δs. Colored: event 201, Δs = 52 μm, Δw = 0.470 cm (1 of 1); EW: event 301, Δs = 73 μm, Δw = 0.079 cm (6 of 10). Paper example (colored): Δs = 83 μm, Δw = 0.074 cm.

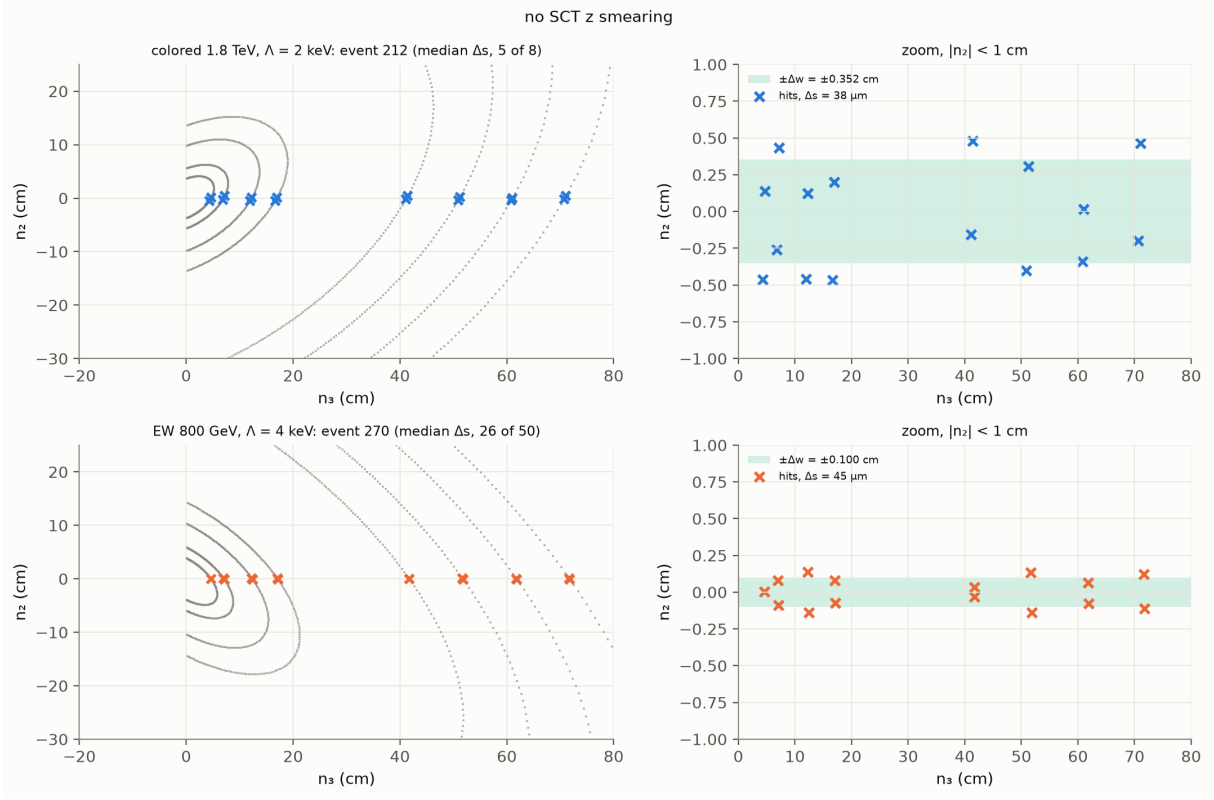


Fig. 4, no SCT z smearing. Same rule. Colored: event 212, Δs = 38 μm, Δw = 0.352 cm (5 of 8); EW: event 270, Δs = 45 μm, Δw = 0.100 cm (26 of 50).

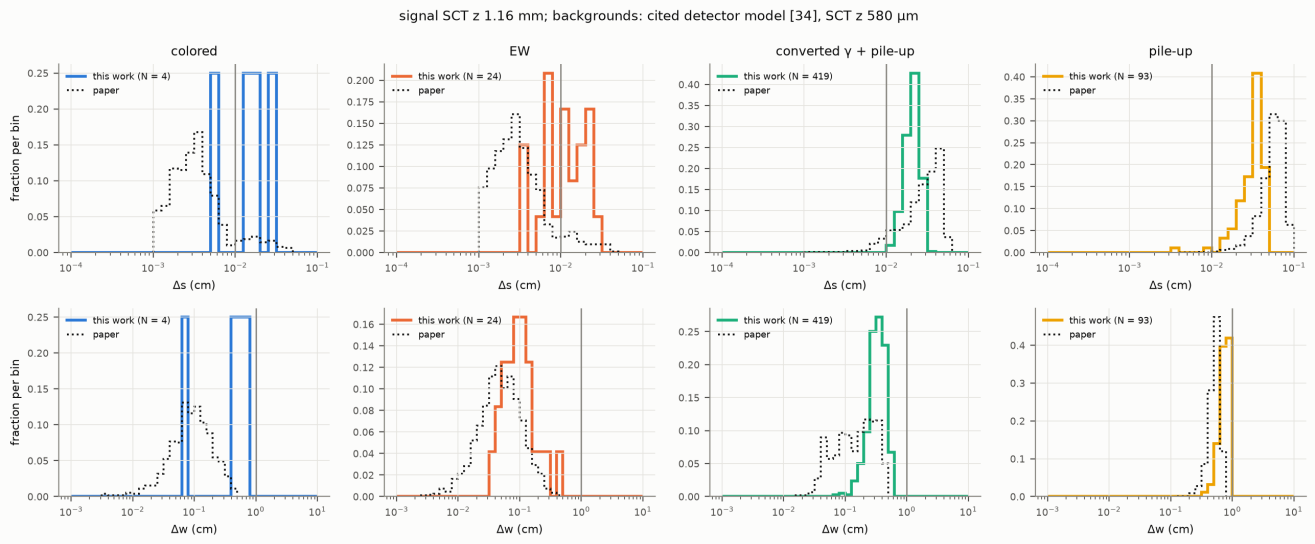


Fig. 5, signal with SCT z 1.16 mm (after plane and hit cuts, before the Δs cut); backgrounds with the [34] model. Dotted: paper.

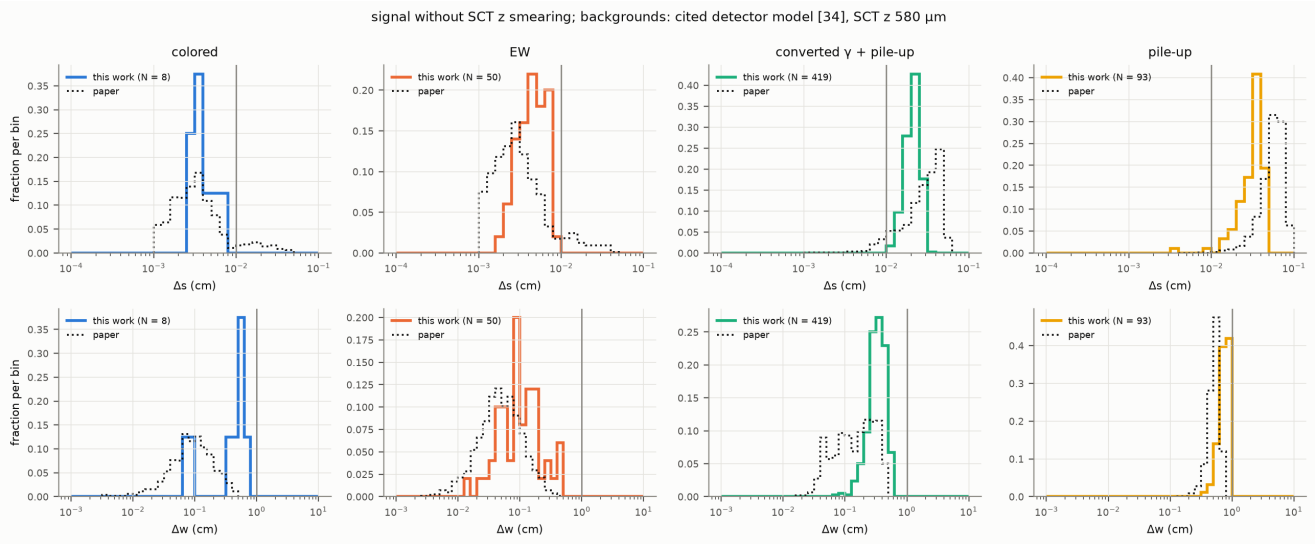


Fig. 5, signal without SCT z smearing; same backgrounds. Dotted: paper.

Fig. 5 medians

Population	Paper Δs	Δs, 1.16 mm	Δs, no smear	Paper Δw	Δw, 1.16 mm	Δw, no smear
colored	32 µm	160 µm (N 4)	38 µm (N 8)	0.092 cm	0.550 cm	0.482 cm
EW	28 µm	110 µm (N 24)	44 µm (N 50)	0.048 cm	0.092 cm	0.097 cm
converted γ + pile-up	306 µm	212 µm (N 419)	212 µm (N 419)	0.156 cm	0.340 cm	0.340 cm
pile-up	571 µm	340 µm (N 93)	340 µm (N 93)	0.525 cm	0.751 cm	0.751 cm

5. Likely reasons for the remaining differences

- **EW trigger (0.18 vs 0.10).** The trigger depends only on the generated pT(QQ̄) spectrum. The paper does not give its matching scale, PDF, scales or Pythia settings. The earlier versions MG5 2.3.3 + Pythia 8.212 named in the paper's source give 0.167, and more than 30 single-setting variations stay at 0.13–0.19. A commented-out draft of Table I in the paper's source lists 0.03; every published version shows 0.10. The colored trigger agrees, so the difference is specific to the EW sample.
- **Reconstruction with SCT z 1.16 mm.** The paper's text gives this resolution, but its own Fig. 5 (signal $\Delta s \approx 30$ μm) and its Λ column are only reproduced without SCT z smearing. This suggests that the signal simulation applied the doubling to R-φ only, or did not smear z.
- **EW ε_{reco} at low Λ** (2 keV: paper 0.41) is higher in our simulation. The paper says ε_{reco} drops at low Λ because of the Δw requirement; our pairs pass it more often.
- **Signal strip width Δw** is about 2× (EW) and 5× (colored) the paper's Fig. 5. Our amplitude follows the paper's Eq. 4 exactly and its Fig. 2 scale. Plotting $\frac{1}{2}\sqrt{\lambda_2}$ would match EW but not colored; the paper's colored Δw is matched by our events near $\Lambda \approx 5$ keV. The cause is not identified.
- **Multiple layer crossings.** Quirks often cross a layer more than once within 25 ns (pair velocity ≈ 0.1–0.2 c against an internal velocity ≈ 0.5 c). The paper expects one crossing; with every crossing as a hit, the reading of “2 hits” (exactly or at least) changes ε_{reco}.
- **Pile-up 4-of-8 rate** (6.6×10^{-3} vs $\sim10^{-3}$) depends mostly on the minimum-bias sample: an inelastic sample gives 2.4×10^{-3}. The paper does not state it.

Questions only the authors can answer: the MG5/Pythia versions and matching settings; the SCT coordinate that was doubled; the Λ of the Fig. 5 signal histograms and the Δw convention plotted; the minimum-bias process and tune used for pile-up.

Statistics

Wilson 95% interval for k of n ($\varepsilon = k/n$, $z = 1.96$): $[\varepsilon + z^2/2n \pm z\sqrt{(\varepsilon(1-\varepsilon)/n + z^2/4n^2)}] / (1 + z^2/n)$. p: exact two-sided binomial probability of a count as unlikely as ours if the paper's value were exact. 95% upper limit for 0 of n: $1 - 0.05^{1/n} \approx 3/n$.

References

[paper] S. Knapen, H. K. Lou, M. Papucci, J. Setford, Phys. Rev. D 96, 115015 (2017), arXiv:1708.02243 (v1 and v2 source compared; Table I identical).
[27] ATLAS, “The Expected Performance of the ATLAS Inner Detector”, ATL-PHYS-PUB-2009-002, in arXiv:0901.0512.
[34] S. Knapen, S. Pagan Griso, M. Papucci, D. J. Robinson, “Triggering soft bombs at the LHC”, arXiv:1612.00850, App. A.
MadGraph5_aMC@NLO: J. Alwall et al., JHEP 07 (2014) 079. Pythia 8: C. Bierlich et al., SciPost Phys. Codeb. 8 (2022).